

RL HW 9

Aditya D and Chris H

Homework 9 Answers

Question 1

- **Conv1:** $3 \times 64 \times 5 \times 5 + 64 = 4,864$
- **Conv2:** $64 \times 128 \times 3 \times 3 + 128 = 73,856$
- **Conv3:** $128 \times 256 \times 3 \times 3 + 256 = 295,168$
- **Conv4:** $256 \times 256 \times 3 \times 3 + 256 = 590,080$
- **Conv5:** $256 \times 128 \times 3 \times 3 + 128 = 295,040$
- **FC1:** $8960 \times 256 + 256 = 2,294,016$
- **FC2:** $256 \times 128 + 128 = 32,896$
- **FC3:** $128 \times 11 + 11 = 1,419$

Total Parameters: 3,587,339

Question 2

The original CNN_small had only 1 convolutional layer and was too shallow to learn complex features, which is why it had poor test accuracy . Key improvements:

- added 2 more convolutional layers (3 total) for hierarchical feature learning
- added batch normalization after each conv layer for stable training
- added max pooling to reduce overfitting
- added dropout (0.5) in fully connected layers
- used data augmentation (RandomCrop, RandomHorizontalFlip)
- applied weight decay (5e-4) and learning rate scheduling

These changes improved test accuracy to 87.49%

Instructions on how to run code and details about the output are in the README